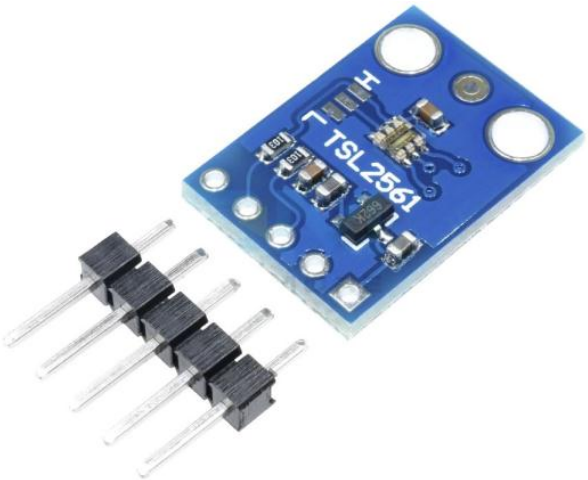
TSL2561 Light Luminosity Sensor Module

The TSL2561 luminosity sensor is an advanced digital light sensor, ideal for use in a wide range of light measurement. Compared to low cost CdS cells, this sensor is more precise, allowing for exact Lux calculations and can be configured for different gain/timing ranges to detect light ranges from up to 0.1 - 40,000+ Lux on the fly. The best part of this sensor is that it contains both infrared and full spectrum diodes! That means you can seperately measure infrared, full-spectrum or human-visible light.

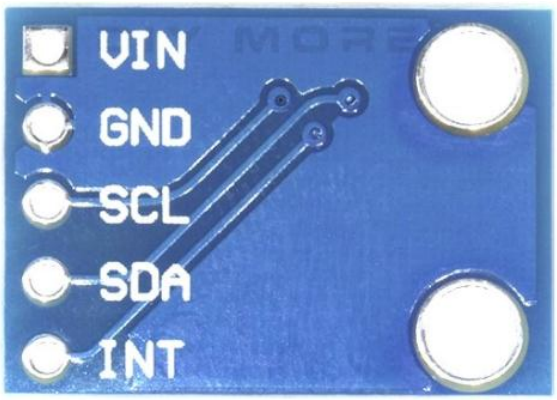


SKU: SSR1026

Description:

- Operating Voltage: 5Vdc.
- Operating Current: 0.6 mA (Max).
- Interface Type: I2C Bus.
- Measurement Range: 0.1 ~ 40,000 Lux.
- Dimensions: (18.2 x 13.2 x 2.5) mm.

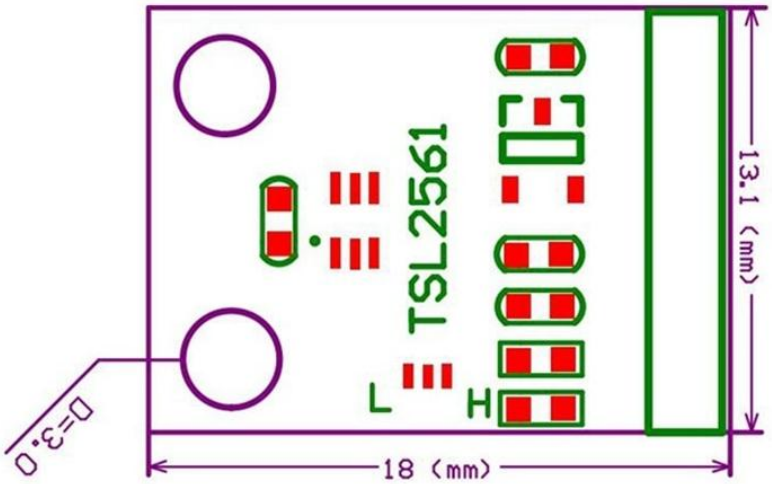
Connection Pins Assignment:



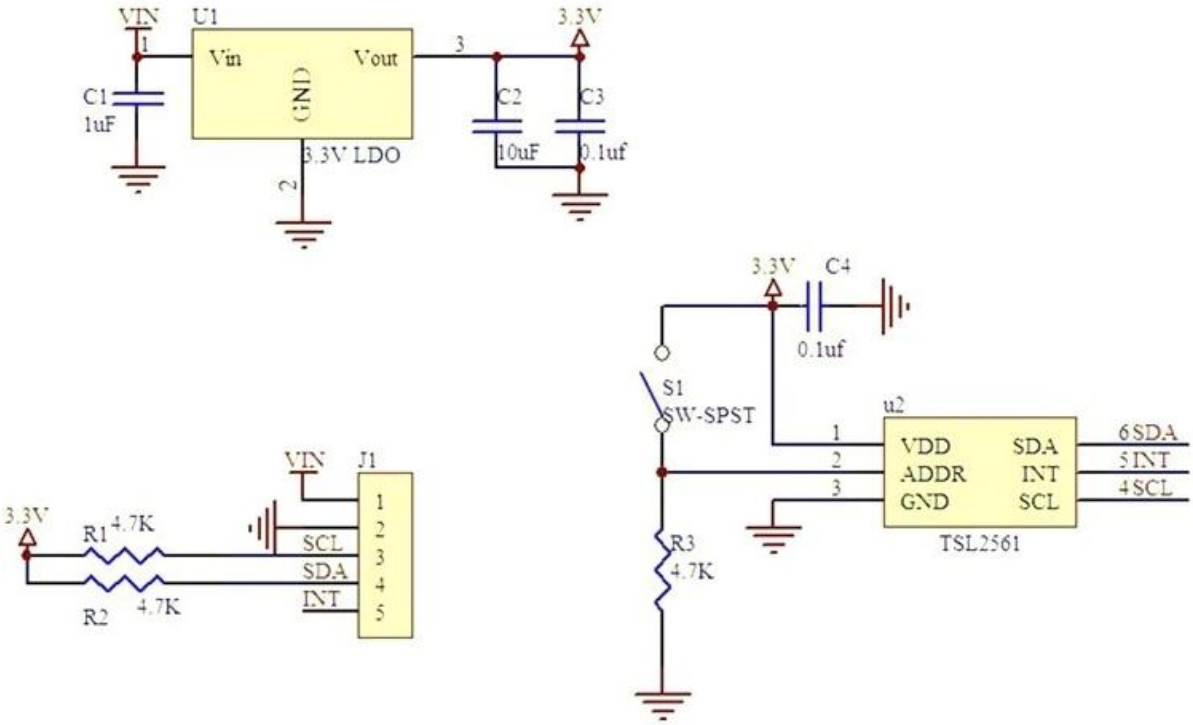
Pin	Function
Vin	Supply input 5Vdc
GND	Ground
SCL	I2C Clock
SDA	I2C Data
INT	Interrupt

Mechanical Dimension:

Unit: mm

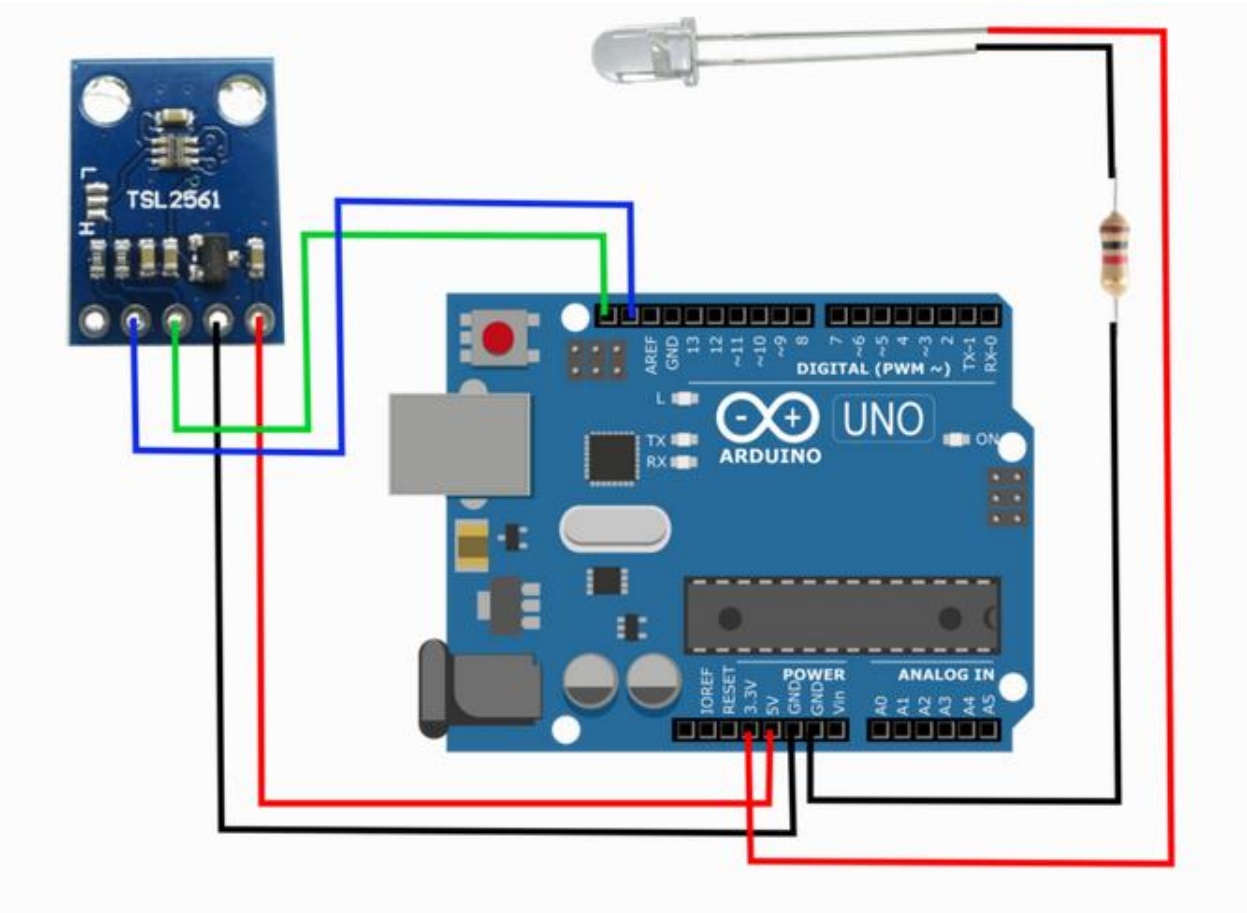


Schematic:


VIN
U1
Vin
Vout
GND
3.3V LDO
C1
1uF
C2
10uF
C3
0.1uf
3.3V
C4
0.1uf
S1
SW-SPST
u2
VDD
ADDR
GND
SDA
INT
SCL
TSL2561
R3
4.7K
J1
SCL
SDA
INT
R1
4.7K
R2
4.7K

Application Examples with Arduino:

Wiring up the TSL2561 module to Arduino Uno as shown in below circuit:



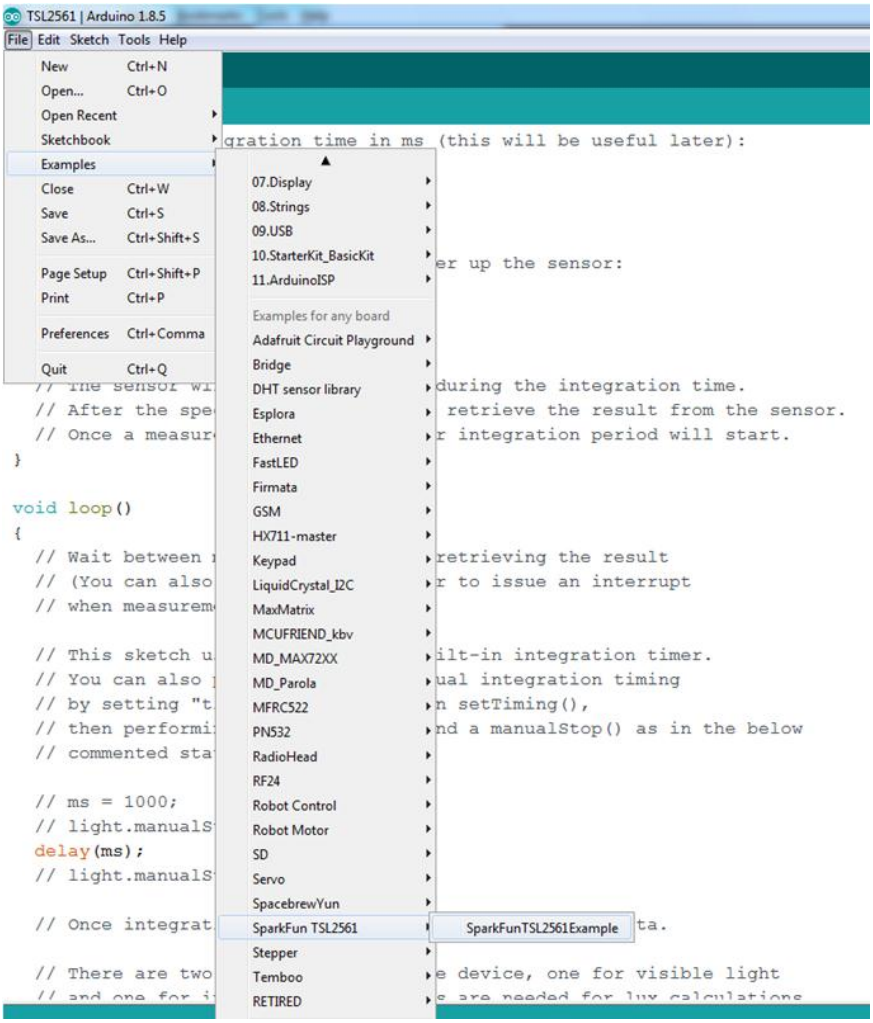
TSL2561	Arduino Uno
Vcc	5V
GND	GND
SCL	A5
SDA	A4

We will be using Spackfun library for this sensor to work in Arduino board. Download the library from the link below and copy to Arduino Library folder.

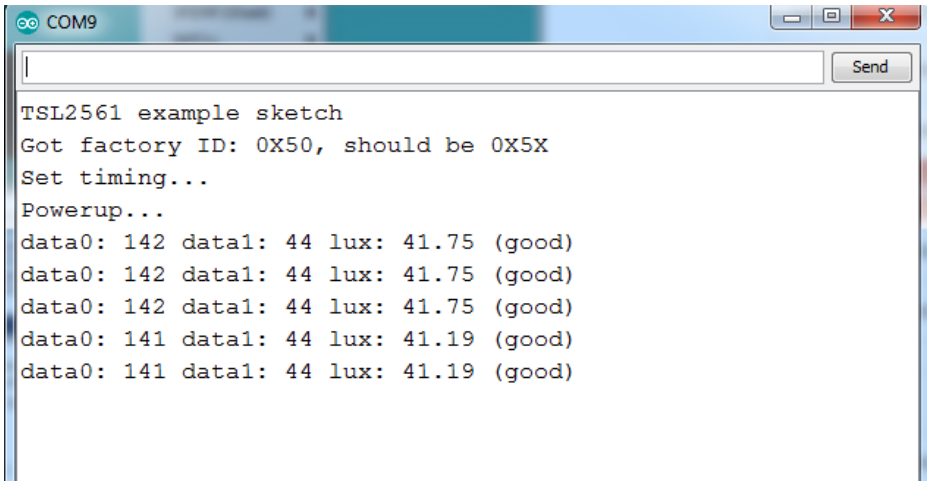
https://github.com/sparkfun/TSL2561_Luminosity_Sensor_BOB

Running the example sketch

The library you just installed includes an example sketch that shows the basic operation of the TSL2561. After you install the library, run the Arduino IDE, and open the following menu item: File / Examples / SparkFun TSL2561 / SparkFunTSL2561example.



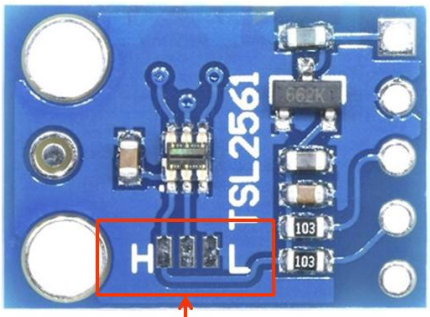
When the example opens, upload it to your Arduino and open the Serial Monitor to 9600 baud. You should see some diagnostic information (if it can't find the device, double check your hardware connections) followed by raw data and lux readings. (See the first page of this tutorial for information on lux.)



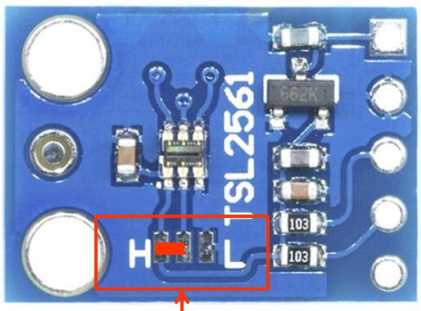
Setting of I2C Bus Slave Address:

The default I2C slave address for TSL2561 sensor is 0x39. This address can be changed by shorting the solder pad , the solder pad location as shown below:

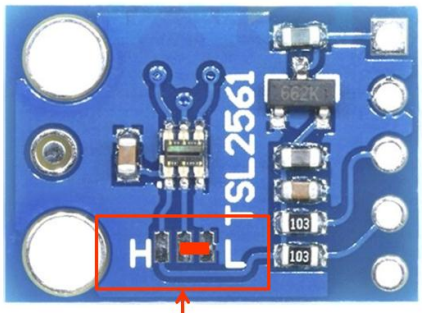
```
light.begin(0x39);  // I2C bus address set to default 0x39,  refer to diagram below  for the other address setting.
```



I2C Slave address setting solder pad



Shorting the center pad to "H" pad, I2C address set to 0x49.



Shorting the center pad to "L" pad, I2C address set to 0x29.

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